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Adoption and content of key audit matters and stock price crash risk

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ABSTRACT

We examine whether the mandate for auditors to report key audit matters (KAMs) affects firm-specific stock price crash risk in China. Auditors in China are required to issue an expanded audit report that contains KAMs for AH-share firms, effective on January 1, 2017 (applicable to the financial year 2016), and for A-share firms, effective on January 1, 2018 (applicable to the financial year 2017). Applying a staggered difference-in-differences (D-i-D) design on a sample of 18,751 observations for financial years 2012–2019, we find that auditor reporting of KAMs is not significantly associated with stock price crash risk. The mechanism tests show that the disclosure of KAMs does not reduce information opaqueness and managerial opportunistic behaviors. Furthermore, our findings are not sensitive, but are robust to firms' corporate governance, product market competition, ownership structure, and auditor size. Overall, our study informs regulators, investors, auditors, and other stakeholders interested in the economic consequences of mandating KAM disclosures.

1. Introduction

Against the backdrop of the controversies surrounding the disclosure of significant audit matters in the new expanded audit report, we examine the impact of this mandatory disclosure on firms' stock price crash risk. Our setting is one of the largest economies, China, which implemented a two-phase compliance strategy with a new audit report (Chinese Institute of Certified Public Accountants (CICPA), 2016). Auditors of all AH-share and A-share firms are required to comply with the new audit reporting rule from January 1, 2017 and January 1, 2018, respectively.[1] These effective dates translate to audits of the 2016 and 2017 financial statements, respectively, because the financial year ends on December 31 for all mainland Chinese headquartered and listed firms. In response to repeated and concerted calls from various stakeholders around the world for more informative and transparent audit reports, regulators in over 60 jurisdictions worldwide implemented an unprecedented mandatory audit report that goes beyond auditors' describing their general audit approach (Financial Reporting Council (FRC), 2013; International Auditing and Assurance Standards Board (IAASB), 2015; Public Company Accounting Oversight Board (PCAOB), 2017). This historic change requires auditors to report specific private information about audit issues and the most significant matters they encountered while conducting an audit. These significant audit issues or matters are known as critical audit matters (hereafter CAMs) in the United States (U.S.) (Public Company Accounting Oversight Board (PCAOB), 2017) or key audit matters (hereafter KAMs) in the International Standards on Auditing (ISA) 701 (International Auditing and Assurance Standards Board (IAASB), 2015). Regulators believe that the disclosure of KAMs can improve the informativeness and transparency of audited financial statements (FRC,2013; International Auditing and Assurance Standards Board (IAASB), 2015; Public Company Accounting Oversight Board (PCAOB), 2017). Some countries, such as the U.S. and China, also require auditors to disclose in the audit report how they addressed the significant matters encountered during the audit.

However, requiring auditors to disclose private information about the significant or critical matters encountered during the audit that are discussed and deliberated with clients' audit committee and management can incur unintended economic consequences to both auditors and

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[1] AH-share firms are listed on both the stock exchanges in mainland China and Hong Kong. A-share firms are listed in mainland China only. Nevertheless, both AH-share and A-share firms are incorporated in mainland China and subject to the same legal and stock exchange requirements.

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clients (Backof, Bowlin, & Goodson, 2022; Chalmers, 2013; Gimbar, Hansen, & Ozlanski, 2016; Reid, Carcello, Li, & Neal, 2019).[2] Many audit practitioners and corporate executives have expressed concerns that the reporting of CAMs/KAMs would merely duplicate existing private communications with the audit client, cause disclosure overload and ambiguity due to repetitive boilerplate language, and increase litigation risk for both auditors and clients (Ernst & Young LLP, 2016; Public Company Accounting Oversight Board (PCAOB), 2017).

This study investigates the impact of the disclosure of KAMs on firm-specific stock price crash risk in China. We motivate our study from the following three perspectives. First, our study responds to the call from regulators around the world, particularly those in developing countries, for more evidence on the economic consequences of the new audit reporting standards (International Federation of Accountants (IFAC), 2016; Independent Regulatory Board for Auditors (IRBA), 2019; Ministry of Finance of the People's Republic of China (MOF), 2019). Unlike in developed countries, corporate governance and investor protection in China and developing countries remain weak (Cheng, Wang, Chiao, Yao, & Fang, 2021; Ke, Lennox, & Xin, 2015; Li & Cai, 2016; Piotroski & Wong, 2012). Furthermore, prior corporate disclosure literature suggests that credible disclosures are helpful to unsophisticated investors, particularly to those in less developed and protected capital markets (Elliott, Krische, & Peecher, 2010; Hirshleifer & Teoh, 2003). Thus, the independent auditor's reporting of previously private information can play an important role in China.

Second, emerging studies provide ambiguous evidence on the capital market's reaction to CAMs/KAMs (Goh, Li, & Wang, 2019; Gutierrez, Minutti-Meza, Tatum, & Vulcheva, 2018; Lennox, Schmidt, & Thompson, 2022; Liao, Minutti-Meza, Zhang, & Zou, 2019; Reid et al., 2019). Distinct from this emerging literature, we focus on firm-specific stock price crash risk because it captures a greater expanse of managerial bad news hoarding activities. According to the bad news hoarding theory, managers strategically withhold bad news and accelerate the release of good news (Hutton, Marcus, & Tehranian, 2009; Jin & Myers, 2006). When this asymmetric disclosure behavior accumulates to a certain point and triggers the sudden release of negative information, the stock price experiences a precipitous drop, resulting in a stock crash (Hutton et al., 2009). Stock price crash risk is of interest to regulators and investors because it captures higher moments of the stock return distribution, that is, extremely negative returns that have considerable economic and market consequences (Callen & Fang, 2015a). Investors are concerned not only with systematic market risk, but also with firm-specific tail risk, as evidenced by the effect of skewness on pricing and implied stock volatility. Therefore, stock price crash risk is critical for portfolio management (Hutton et al., 2009; Robin & Zhang, 2015). Research on stock price crash risk has gained momentum since this recognition and provides a relevant context to examine KAM disclosures.[3]

Third, the staggered adoption of the expanded audit report based on firm size or listing attributes is currently unique to China and the U.S., but China adopted it much sooner. Auditors of non-large accelerated filers in the U.S. did not have to comply with the new reporting standard until December 2020 (Public Company Accounting Oversight Board (PCAOB), 2017), which precludes researchers from examining longer horizon market reactions to this unprecedented audit regulation. We exploit China's full adoption setting to employ a robust difference-in-differences (D-i-D) design to examine the economic consequences of the new audit report without potential confounding factors from having to employ a control sample from a different country (Ke et al., 2015).

We posit that the KAMs disclosed in the expanded audit report may or may not affect stock price crash risk. As we develop our hypotheses later, we present a summary of the arguments on the relation between the disclosure of KAMs and stock price crash risk. First, auditors have strong reputational and litigation risk incentives to closely monitor management's bad news hoarding activities, thus improving the credibility of audited information (Khurana & Raman, 2004; Robin & Zhang, 2015). Second, disclosure of KAMs improves corporate transparency to investors and reveals significant risks, particularly risks boarding by managers, which may attract greater market scrutiny (International Auditing and Assurance Standards Board (IAASB), 2015). The disclosure of KAMs mitigates the biases in the information environment, and thus reduces stock price crash risk. Third, improved corporate transparency can help mitigate stock price crash risk by reducing the divergence of investment opinions among investors (Hong & Stein, 2003; Hutton et al., 2009; Robin & Zhang, 2015).

However, auditor reporting of KAMs may arguably have no impact on stock price crash risk for three important reasons. Research finds that (1) KAMs do not convey new information (Files & Gencer, 2020; Gutierrez et al., 2018; Lennox et al., 2022; Liao et al., 2019), (2) standardized/boilerplate KAM disclosures are too generic to reveal client-specific risks (Brasel, Doxey, Grenier, & Reffett, 2016; Public Company Accounting Oversight Board (PCAOB), 2017), and (3) auditors may strategically avoid communicating areas of significant client risk to the public in order to reduce potential litigation risk (Rousseau & Zehms, 2020). These competing views suggest that the proposed benefits of mandating KAM disclosures remain an open empirical question.

To perform the staggered D-i-D analysis, our sample comprises all AH-share and A-share firms with available data for the financial years 2012–2019. This sample period spans up to four years prior to and three years following the first financial year (i.e., 2016) for which auditors are required to report KAMs. Consistent with prior studies (Chen, Hong, & Stein, 2001; Kim, Li, & Zhang, 2011a, 2011b), we use two well-established measures to proxy for firm-specific stock price crash risk, (1) the negative skewness of future firm-specific weekly returns, and (2) the asymmetric volatilities between negative and positive firm-specific weekly returns. In the robustness test, we also apply a dummy variable to measure the likelihood of future crash risk by counting if a firm ever experiences one or more crash weeks in a given year.

The staggered D-i-D results indicate no significant difference in stock price crash risk before and after the adoption of the new audit report requirements. Our staggered D-i-D satisfies the parallel trends assumption, and our finding is also not sensitive to alternative measure of crash risk and different stock price crash risk horizon windows. When we examine the nature and complexity of KAM disclosures (i.e., the number and length of KAMs and auditors' responses) and their information content based on textual analysis, we find that KAMs and auditors' responses have no impact on crash risk. We then perform two mechanism tests to examine whether corporate information opaqueness and managerial opportunism could be potential mechanisms, and our results suggest that the mediating effects of corporate information opaqueness and managerial opportunism are insignificant. Further, we perform several heterogeneity analyses to demonstrate that the insignificant association between the adoption of KAMs and stock price crash risk is not conditional on the strength of corporate governance, market competition, ownership structure or auditor size. To alleviate the concern that insignificant results could be due to low power, we estimate the statistical power of all our regressions and find that the power exceeds the 0.80 threshold (Cohen, 1988). Therefore, our consistent, robust, and statistically powerful results imply that mandating auditors to report KAMs does not improve corporate transparency, nor alter management bad news hoarding.

Our study makes the following contributions to the literature: First, we extend and complement emerging research on the disclosure of

[2] Some of the economic consequences cited in this literature include higher audit costs, audit delay due to communication with the audit committee and client management, greater litigation risk, risk of proprietary information spillover to competitors, and delay in the filing of financial statements.

[3] For instance, see Callen and Fang (2017), Chang, Chen, and Zolotoy (2017), Chen, Kim, Li, and Liang (2018), Cheng et al. (2021), DeFond et al. (2015), Kim et al. (2016, 2011a, 2011b), and Li and Zhan (2019).

Table 1
Descriptive statistics.

Panel A: Descriptive statistics of key variables

Variable	N	Mean	Std. Dev	Min	Median	Max
$NCSKEW_t$	18,751	−0.286	0.718	−2.382	−0.248	1.674
$DUVOL_t$	18,751	−0.184	0.488	−1.377	−0.185	1.074
$NCSKEW_{t-1}$	18,751	−0.268	0.709	−2.332	−0.233	1.712
$DUVOL_{t-1}$	18,751	−0.168	0.473	−1.303	−0.169	1.073
DISKAM	18,751	0.442	0.497	0.000	0.000	1.000
$DTURN_{t-1}$	18,751	−0.029	0.234	−0.662	−0.029	0.690
$SIGMA_{t-1}$	18,751	0.049	0.020	0.017	0.045	0.115
RET_{t-1}	18,751	−0.136	0.120	−0.659	−0.098	−0.014
$SIZE_{t-1}$	18,751	22.170	1.284	19.710	22.000	26.090
ROA_{t-1}	18,751	0.037	0.055	−0.216	0.035	0.187
$LOSS_{t-1}$	18,751	0.093	0.291	0.000	0.000	1.000
REC_{t-1}	18,751	0.117	0.104	0.000	0.093	0.466
INV_{t-1}	18,751	0.152	0.144	0.000	0.116	0.737
LEV_{t-1}	18,751	0.434	0.211	0.052	0.428	0.902
MB_{t-1}	18,751	3.857	3.787	0.726	2.767	28.110
$DACC_{t-1}$	18,751	0.067	0.074	0.001	0.044	0.421
SOE_{t-1}	18,751	0.386	0.487	0.000	0.000	1.000
SHR_{t-1}	18,751	34.810	14.990	8.505	32.830	74.980
$BIG6_{t-1}$	18,751	0.296	0.456	0.000	0.000	1.000
INS_{t-1}	18,751	0.373	0.238	0.000	0.376	0.877
ANA_{t-1}	18,751	1.515	1.154	0.000	1.609	3.738

Panel B: Descriptive statistic of KAMs content

Variable	N	Mean	Std. Dev	Min	Median	Max
KAMNUM (Raw)	8197	2.084	0.652	1.000	2.000	4.000
KAMNUM (Log)	8197	0.682	0.337	0.000	0.693	1.386
KAMWRDS (Raw)	8197	415.000	202.100	87.000	383.000	1131.000
KAMWRDS (Log)	8197	5.905	0.516	4.466	5.948	7.031
RESPNUM (Raw)	8197	11.250	4.860	2.000	11.000	26.000
RESPNUM (Log)	8197	2.308	0.515	0.693	2.398	3.258
RESPWRDS (Raw)	8197	594.300	243.800	164.000	562.000	1381.000
RESPWRDS (Log)	8197	6.301	0.427	5.100	6.332	7.231

Panel C: Classification of KAM topics

KAM Category	Number	Percentage
Revenue recognition	5334	31.064
Accounts receivable allowance and Loan impairment	2917	16.988
Intangible assets and goodwill impairment	2779	16.184
Inventory	1785	10.395
Property, plant, and equipment measurement and impairment	937	5.457
Investments	468	2.726
M&As	401	2.335
Related-party transactions	302	1.759
Fair value and financial instrument	220	1.281
TOTAL	17,171	100

Notes: This table presents the sample descriptive statistics. Panel A presents descriptive statistics of the variables used in the regression analyses. Panel B presents the statistics of KAM content for 8197 observations that disclose KAMs in the expanded audit reports. Panel C presents the distribution of the KAM category. All variables are defined in Appendix B.

KAMs. Specifically, we contribute to the ongoing discussion by providing novel evidence on the consequences of KAM disclosure from the perspective of stock price crash risk. To clarify the inconclusive recent literature that investigates the short-window market reactions to KAMs (Files & Gencer, 2020; Goh et al., 2019; Gutierrez et al., 2018; Lennox et al., 2022; Liao et al., 2019), we distinctively focus on management bad news hoarding captured by stock price crash risk (Kim, Li, & Li, 2014). As such, we provide a fresh perspective on the economic consequences of KAM disclosure made by auditors. Our findings suggest that the unprecedented mandate to disclose KAMs in the new audit report does not reduce managerial bad news hoarding risk.

Second, we extend and complement nascent literature (e.g., Files & Gencer, 2020; Gutierrez et al., 2018; Lennox et al., 2022; Liao et al., 2019) by adopting a design choice that enables us to examine, beyond the KAM disclosure dichotomy, whether the nature and content of the KAM disclosures are important. This set of empirical analysis implies that the disclosure of KAMs does not improve corporate transparency.

Our research informs policy-oriented discussions on the informativeness of the new expanded audit reports. The International Federation of Accountants (IFAC) (2016) highlights the significant evolution of the audit around the world, in particular pertaining to capital market participants demanding auditors to issue more meaningful opinions beyond the "pass/fail" information in audit reports. This pressure led to the development of the ISA 701. Similarly, the PCAOB (2017, 1) emphasizes that traditional audit reports "convey little information", and that the adoption of the new expanded audit report is expected to respond to the investors' demand for improved communication. Our study can inform such policy discussions to the extent that the institutional environment is similar. Given that the disclosure of CAMs/KAMs in the new audit report is relatively uniform across jurisdictions and touted as one of the most significant changes to the audit report, our findings inform policy makers and regulators interested in the consequences of such disclosures (International Auditing and Assurance Standards Board (IAASB), 2015; Independent Regulatory Board for Auditors (IRBA), 2019; Ministry of Finance of the People's Republic of China (MOF), 2019). Pending further research by academics and regulators, we infer that disclosure of KAMs does not improve corporate transparency and suggest that policymakers may need to

Table 2
Pearson correlation matrix.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
1.NCSKEW$_t$	1.000																								
2.DUVOL$_t$	**0.874**	1.000																							
3.NCSKEW$_{t-1}$	**0.052**	**0.046**	1.000																						
4.DUVOL$_{t-1}$	**0.042**	**0.041**	**0.877**	1.000																					
5.DISKAM	**0.073**	**0.069**	**0.027**	**0.041**	1.000																				
6.KAMNUM	0.015	0.007	0.014	0.016	−0.008	1.000																			
7.KAMWRDS	0.018	0.010	0.007	0.005	**−0.021**	**0.707**	1.000																		
8.RESPNUM	**0.026**	0.016	**0.032**	**0.036**	0.004	**0.624**	**0.501**	1.000																	
9.RESPWRDS	0.010	−0.007	**0.039**	**0.036**	−0.016	**0.750**	**0.645**	**0.645**	1.000																
10.DTURN$_{t-1}$	**−0.045**	**−0.052**	0.004	−0.010	**−0.359**	−0.001	−0.016	−0.015	**0.035**	1.000															
11.SIGMA$_{t-1}$	**0.023**	**0.017**	**−0.031**	**−0.018**	**−0.165**	0.016	**−0.032**	0.008	**0.037**	**0.449**	1.000														
12.RET$_{t-1}$	**−0.013**	−0.009	**0.053**	**0.039**	**0.164**	−0.009	**0.038**	−0.001	**−0.027**	**−0.436**	**−0.972**	1.000													
13.SIZE$_{t-1}$	**−0.069**	**−0.085**	**−0.085**	**−0.107**	**0.127**	**0.119**	**0.171**	**0.046**	**0.095**	**0.012**	**−0.275**	**0.234**	1.000												
14.ROA$_{t-1}$	**0.039**	**0.036**	0.002	−0.005	−0.009	**−0.097**	**−0.053**	**−0.026**	**−0.048**	**−0.063**	**−0.125**	**0.125**	−0.009	1.000											
15.LOSS$_{t-1}$	0.007	0.007	**0.040**	**0.045**	−0.007	**0.048**	0.007	**0.021**	**0.022**	**0.071**	**0.124**	**−0.118**	**−0.062**	**−0.644**	1.000										
16.REC$_{t-1}$	**0.043**	**0.048**	**0.054**	**0.067**	**0.083**	**0.098**	**0.053**	**0.092**	**0.121**	−0.006	**0.139**	**−0.116**	**−0.180**	−0.001	**−0.026**	1.000									
17.INV$_{t-1}$	−0.010	**−0.013**	**−0.023**	**−0.026**	**−0.086**	**0.059**	**0.046**	−0.000	**0.054**	0.005	**−0.030**	**0.036**	**0.128**	**−0.092**	−0.001	**−0.103**	1.000								
18.LEV$_{t-1}$	**−0.056**	**−0.064**	**−0.059**	**−0.069**	**−0.038**	**0.123**	**0.121**	**0.048**	**0.099**	**0.025**	**−0.064**	**0.056**	**0.504**	**−0.367**	**0.180**	**−0.020**	**0.318**	1.000							
19.MB$_{t-1}$	**0.073**	**0.071**	**0.087**	**0.079**	**−0.061**	**−0.021**	**−0.037**	−0.010	**−0.054**	**0.099**	**0.351**	**−0.348**	**−0.371**	**−0.020**	**0.134**	**0.083**	**−0.049**	**0.068**	1.000						
20.DACC$_{t-1}$	**0.016**	**0.016**	**0.042**	**0.033**	**−0.097**	**0.023**	0.005	−0.015	**0.018**	−0.007	**0.089**	**−0.080**	**−0.084**	**−0.109**	**0.116**	**0.016**	**0.140**	**0.105**	**0.164**	1.000					
21.SOE$_{t-1}$	**−0.096**	**−0.098**	**−0.108**	**−0.115**	**−0.087**	**−0.071**	**−0.044**	**−0.102**	**−0.108**	**0.020**	**−0.182**	**0.159**	**0.350**	**−0.095**	**0.036**	**−0.200**	**0.044**	**0.304**	**−0.129**	**−0.042**	1.000				
22.SHR$_{t-1}$	**−0.038**	**−0.042**	**−0.051**	**−0.058**	**−0.069**	**−0.059**	**−0.026**	**−0.044**	**−0.041**	0.005	**−0.089**	**0.074**	**0.229**	**0.122**	**−0.081**	**−0.102**	**0.050**	**0.061**	**−0.110**	**−0.023**	**0.228**	1.000			
23.BIG6$_{t-1}$	**−0.024**	**−0.024**	**−0.031**	**−0.027**	**0.042**	**−0.028**	**0.066**	**−0.064**	**−0.066**	**0.029**	−0.010	0.005	**0.165**	**0.041**	**−0.024**	−0.004	**−0.042**	**0.030**	−0.007	**−0.036**	**0.067**	**0.081**	1.000		
24.INS$_{t-1}$	**−0.037**	**−0.049**	**−0.037**	**−0.055**	−0.009	−0.006	**0.033**	**−0.031**	**−0.034**	**0.073**	**−0.167**	**0.146**	**0.443**	**0.061**	**−0.035**	**−0.164**	**0.040**	**0.228**	**−0.098**	**−0.053**	**0.384**	**0.304**	**0.078**	1.000	
25.ANA$_{t-1}$	**0.047**	**0.034**	**0.058**	**0.038**	**−0.030**	**0.054**	**0.094**	**0.043**	**0.044**	0.003	**−0.128**	**0.120**	**0.368**	**0.393**	**−0.218**	0.011	**−0.051**	**−0.049**	**−0.071**	**−0.050**	**−0.045**	**0.093**	**0.066**	**0.266**	1.000

Notes: This table presents the Pearson correlation matrix. Bold values indicate statistical significance at the 10%, 5% or 1% level. All variables are defined in Appendix B.

Table 3
Regressions of stock price crash risk on adoption of KAM audit standard.

	(1)	(2)	(3)	(4)
Dependent variables =	NCSKEW	DUVOL	NCSKEW	DUVOL
	Coeff.	Coeff.	Coeff.	Coeff.
	(t-stat.)	(t-stat.)	(t-stat.)	(t-stat.)
DISKAM	**−0.131**	**−0.077**	**−0.093**	**−0.051**
	(−1.203)	**(−1.114)**	**(−0.861)**	**(−0.731)**
$NCSKEW_{t-1}$			−0.125***	
			(−14.633)	
$DUVOL_{t-1}$				−0.134***
				(−15.252)
$DTURN_{t-1}$			−0.013	−0.030
			(−0.408)	(−1.355)
$SIGMA_{t-1}$			4.710***	2.597***
			(3.247)	(2.595)
RET_{t-1}			0.441**	0.232
			(1.965)	(1.480)
$SIZE_{t-1}$			0.103***	0.056***
			(4.520)	(3.591)
ROA_{t-1}			0.513**	0.300**
			(2.432)	(2.065)
$LOSS_{t-1}$			0.101***	0.058***
			(3.211)	(2.761)
REC_{t-1}			0.049	0.104
			(0.292)	(0.888)
INV_{t-1}			−0.025	0.004
			(−0.184)	(0.041)
LEV_{t-1}			−0.314***	−0.200***
			(−3.660)	(−3.444)
MB_{t-1}			0.019***	0.012***
			(5.929)	(5.359)
$DACC_{t-1}$			0.058	0.089
			(0.609)	(1.352)
SOE_{t-1}			−0.082	−0.093*
			(−1.193)	(−1.849)
SHR_{t-1}			0.001	0.000
			(0.680)	(0.192)
$BIG6_{t-1}$			−0.044	−0.021
			(−1.512)	(−1.084)
INS_{t-1}			0.069	0.041
			(1.450)	(1.279)
ANA_{t-1}			0.004	0.009
			(0.358)	(1.207)
Constant	−0.369***	−0.246***	−2.777***	−1.565***
	(−20.582)	(−20.489)	(−5.713)	(−4.683)
Firm and Year FE	Included	Included	Included	Included
Cluster Firm	Included	Included	Included	Included
N	18,751	18,751	18,751	18,751
Adj. R^2	0.059	0.057	0.080	0.079
F-value	39.268***	37.456***	27.468***	27.604***
Statistical power	1.000	1.000	1.000	1.000

Notes: This table reports the results of the staggered difference-in-differences regressions specified in Eq. (4). All variables are defined in Appendix B. All continuous variables are winsorized at the top and bottom 1 %. Robust t-statistics to address heteroscedasticity are reported in parentheses below the respective coefficient. *, **, and *** represent two-tailed significance levels of 0.10, 0.05, and 0.01, respectively.

research the auditors' narrative on KAMs.

The remainder of this paper is structured as follows. Section 2 discusses the context of the study and reviews the literature culminating in our hypothesis. Section 3 discusses the research design and sample. The primary results and robustness tests are reported in Section 4. Section 5 reports additional analyses, and Section 6 concludes the paper.

2. Background, literature review and hypotheses

2.1. The new audit reporting standards

The United Kingdom (UK) was the first country mandating auditors to disclose private information about the risks encountered by auditors during their financial statement audits. The Financial Reporting Council (FRC) issued the International Standard on Auditing (UK and Ireland) 700, which requires auditors to disclose the risks of material misstatement that have the "greatest effect on the overall audit strategy, the allocation of resources in the audit, and directing the efforts of the engagement team" for periods commencing on or after October 1, 2012 (Financial Reporting Council (FRC), 2013, 6). Auditors should also disclose "an explanation of how the auditor applied the concept of materiality in planning and performing the audit" (Financial Reporting Council (FRC), 2013, 7). In the U.S., the Public Company Accounting Oversight Board (PCAOB) introduced a similar auditing standard requiring auditors to communicate CAMs for large accelerated filer audits effective as of June 30, 2019, and for all other filers effective from December 15, 2020 (Public Company Accounting Oversight Board (PCAOB), 2017).

At a broader international level, the International Auditing and Assurance Standards Board (IAASB) issued a similar auditor reporting standard, ISA 701, which took effect on December 15, 2016. ISA 701 requires auditors to communicate additional audit information about matters that are of the most significance during the audit, where the significance of a matter is based on the auditor's professional judgment. The ISA 701 also requires the auditor to describe the most relevant response or approach taken to address the identified KAMs.

2.2. The new auditing requirements in China

The auditing profession in China emerged and started to develop in the 1980s. The Company Law of 2013 and relevant securities regulations established independent external audit requirements for all listed firms. The Chinese Institute of Certified Public Accountants (CICPA) has the legal authority to draft Chinese auditing standards (CSAs), which have been adopted since January 1, 2007. To promote the comprehensive convergence of CSAs and international auditing standards, the Ministry of Finance of The People's Republic of China (MOF) issued Certified Public Accountants Auditing Standards No. 1504 (CSA No. 1504) in 2016 (the Ministry of Finance of the People's Republic of China (MOF), 2016), which requires auditors to communicate KAMs in the audit report. KAMs are defined as matters that are of most significance in the process of auditing the current financial reports based on the auditor's professional judgment (Article 7).

More specifically, when determining whether a matter is critical and requires disclosure as a KAM, auditors need to consider three factors: (1) areas where the assessed risk of material misstatement is higher, (2) significant audit judgments related to areas involving crucial management judgments and estimates in the financial statements, and (3) the impact of major transactions or events in the current period (Article 9). When disclosing KAMs,[4] auditors are required to include a separate section in the audit report, with the Key Audit Matters as the title, and use appropriate subheadings to describe each specific KAM (Article 11). Articles 11 and 13 further require auditors to describe why the matters were of most significance during the audit. Auditors also need to discuss how these matters were addressed in the audit procedures.

In Appendix A we provide an example of KAMs contained in the audit report of Air China (Stock code: 00753). The KAM reporting requirements and auditor considerations coupled with the example in Appendix A illustrate that the KAM requirements are fairly similar to the

[4] Unless one of the following situations exists, auditors should disclose KAMs in the audit report: (1) Laws and regulations prohibit public disclosure of a certain matter; (2) in very rare cases, if it is reasonably expected that the negative consequences of communicating certain matters in the audit report outweigh the benefits in terms of public interest, auditors determine that the matter should not be communicated in the audit report. If the audited entity has publicly disclosed information related to the matter, this provision does not apply.

Table 4
Regressions of stock price crash risk on KAM content.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Dependent variable=	*NCSKEW*	*DUVOL*	*NCSKEW*	*DUVOL*	*NCSKEW*	*DUVOL*	*NCSKEW*	*DUVOL*	*NCSKEW*	*DUVOL*
Panel A: Regressions of stock price crash risk on the number and word count of KAMs and auditor responses (n = 5035)										
KAMNUM	**0.037**	**0.002**								
	(0.592)	**(0.037)**								
KAMWRDS			**0.006**	**−0.013**						
			(0.143)	**(−0.438)**						
RESPNUM					**0.018**	**−0.002**				
					(0.546)	**(−0.069)**				
RESPWRDS							**−0.002**	**−0.022**		
							(−0.034)	**(−0.624)**		
Controls	Included	Included	Included	Included	Included	Included	Included	Included		
Firm and Year FE	Included	Included	Included	Included	Included	Included	Included	Included		
Cluster Firm	Included	Included	Included	Included	Included	Included	Included	Included		
N	8197	8197	8197	8197	8197	8197	8197	8197		
Adj. R^2	0.158	0.158	0.158	0.158	0.158	0.158	0.158	0.158		
F-value	24.150***	26.003***	24.129***	25.990***	24.128***	25.984***	24.145***	26.005***		
Statistical power	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000		
Panel B: Regressions of stock price crash risk on top-5 KAM topics (n = 5035)										
REVENUE	**−0.011**	**0.004**								
	(−0.260)	**(0.131)**								
RECVLOAN			**−0.024**	**−0.028**						
			(−0.507)	**(−0.833)**						
INTANGIBLE					**0.028**	**0.008**				
					(0.520)	**(0.218)**				
INVENTORY							**−0.028**	**−0.029**		
							(−0.487)	**(−0.734)**		
PPE									**0.017**	**0.018**
									(0.277)	**(0.453)**
Controls	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included
Firm and Year FE	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included
Cluster Firm	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included
N	8197	8197	8197	8197	8197	8197	8197	8197	8197	8197
Adj. R^2	0.158	0.158	0.158	0.158	0.158	0.158	0.158	0.158	0.158	0.158
F-value	24.150***	26.018***	24.139***	26.067***	24.129***	25.969***	24.162***	25.984***	24.146***	26.006***
Statistical power	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Notes: This table presents regressions of stock price crash risk on the content of the KAM contained in the audit report for the KAM audit standard adoption period. Panel A presents regressions related to the quantity of KAMs and auditors' responses to the KAMs. Panel B presents regressions of the top-5 KAM topics reported in Table 1. Each of the top-5 KAM topic indicator variables equal to 1 if the auditor reports the specific topic as a KAM, and 0 otherwise. All other variables are defined in Appendix B. All continuous variables are winsorized at the top and bottom 1 %. All control variables, and firm and year fixed effects are included in the regressions but not tabulated for brevity. Robust t-statistics to address heteroscedasticity are reported in parentheses below the respective coefficient. *, **, and *** represent two-tailed significance levels of 0.10, 0.05, and 0.01, respectively.

CAM/KAM requirements in other countries. This uniformity stems from the convergence of CSAs with the International Auditing Standards, which increases the generalizability of our findings.

2.3. *The consequences of expanded audit reports*

Earlier studies examine consequences of the expanded audit reports and find inconclusive results. Using eye-tracking technology, Sirois, Bédard, and Bera (2018) show that the disclosure of KAMs attracts investor attention, but it also distracts them from paying attention to relevant information in the financial statements. Reid et al. (2019) find that compared to U.S. and European firms, UK firms experience a greater reduction in absolute abnormal accruals and the propensity to meet or beat analyst forecasts after the implementation of the new expanded audit report. Reid et al. (2019) also document a significantly higher earnings response coefficient (ERC), suggesting that KAMs improve investors' perceptions of audited financial information. However, other studies find that the expanded audit report provides little incremental information to investors (Gutierrez et al., 2018; Lennox et al., 2022). Gutierrez et al. (2018) examine the cumulative absolute abnormal returns and the sum of three-day abnormal trading volumes around the new audit report filing date, and find no evidence that investors value the information disclosed in the new audit report in the UK. They reason that KAM disclosures do not reliably capture the risks relevant to investors. Lennox et al. (2022) reinforce this conclusion by finding insignificant market reactions to KAM disclosures in UK audit reports. They argue that investors were already informed of the firm's risks through earlier disclosure channels, such as prior earnings announcements, conference calls, and annual reports. With the availability of U.S. data on extended auditor reporting, Files and Gencer (2020) also find an insignificant effect of CAMs on investor responses.

In supplementing studies in the UK and the U.S., three concurrent studies examine the economic consequences of KAMs in Hong Kong and China and find mixed results. Specifically, Liao et al. (2019) show that KAMs are not significantly associated with abnormal returns, abnormal trading volumes, or bid-ask spreads in Hong Kong. They conclude that the expanded audit report does not improve information conveyed to investors, which is contrary to regulators' expectations. By contrast, Goh et al. (2019) examine firms listed in mainland China and show that abnormal trading volume and ERC are higher and stock price synchronicity is lower after the adoption of the expanded audit report. Different from market reaction analyses, Zeng, Zhang, Zhang, and Zhang (2021) find that audit quality improves following the mandatory KAM disclosure rule. We complement this emerging yet inconclusive body of concurrent literature examining mostly short-term market reactions by considering stock price crash risk.

Table 5
Parallel trends assumption analyses.

	(1)	(2)
Dependent Variable =	NCSKEW	DUVOL
T-3	0.090	0.156
	(0.494)	(1.275)
T-2	0.036	0.043
	(0.318)	(0.615)
T	−0.002	−0.012
	(−0.017)	(−0.154)
T + 1	0.058	−0.007
	(0.283)	(−0.056)
T + 2	0.438	0.188
	(1.534)	(1.030)
Controls	Included	Included
Firm and Year FE	Included	Included
Cluster Firm	Included	Included
N	18,678	18,678
Adj. R^2	0.081	0.080
F-value	24.622***	24.582***
Statistical power	1.000	1.000

Notes: This table presents the results of parallel assumption analysis. *T-3* equals one if the financial year is up to three years before the disclosure of KAMs becomes mandatory, and zero otherwise. *T-2* equals one if the financial year is two years before the disclosure of KAMs becomes mandatory, and zero otherwise. *T* equals one for the financial year when the disclosure of KAMs is mandatory, and zero otherwise. *T + 1* is an indicator variable equal to one if the financial year is one year after the mandatory disclosure of KAMs, and zero otherwise. *T + 2* is an indicator variable equal to one if the financial year is two year after the mandatory disclosure of KAMs, and zero otherwise. We use *T-1* as the benchmark. All other variables are defined in Appendix B. All continuous variables are winsorized at the top and bottom 1 %. All control variables are included in the regression but not tabulated for brevity. Robust t-statistics in parentheses. The reported significance is based on robust standard errors of two-tailed tests, adjusted for heteroscedasticity. *, **, and *** represent significance levels of 0.10, 0.05, and 0.01, respectively.

2.4. Stock price crash risk

Under the agency perspective, due to the information asymmetry between managers and investors, managers have certain discretion over when to allow public access of corporate fundamental information, and thus they have incentives to conceal negative news (Kothari, Shu, & Wysocki, 2009). When the negative news is stockpiled to a certain point that managers can no longer withhold, it leads to a sudden release of bad news to the public which causes a stock price crash consequently (Hutton et al., 2009; Jin & Myers, 2006).

Prior literature examining stock price crash risks has provided empirical evidence that supports the bad news hoarding theory. The factors that are associated with crash risks can be categorized as follows: information transparency, managerial behavior, and monitoring and governance. First, studies have shown that stock price crash risks are higher when there is a lack of full transparency concerning firm performance (Hutton et al., 2009; Jin & Myers, 2006). In particular, decreased stock price crash risk is significantly associated with higher level of financial reporting transparency (Kim & Zhang, 2014), accounting conservatism (Kim & Zhang, 2016), financial statement comparability (Kim, Li, Lu, & Yu, 2016), and lower operating cash flow opacity (Cheng, Li, & Zhang, 2020). Second, managerial actions can contribute to stock price crashes. Managers' opportunity to hoard negative news increases crash risk, as observed in cases of higher corporate tax avoidance levels (Kim et al., 2011a). Furthermore, studies indicate that stronger equity-based incentives (Kim et al., 2011b) and competitive pressure from the product market (Li & Zhan, 2019) enhance managers' motivation for hoarding bad news, leading to higher stock crash risks. Third, prior studies also document that several monitoring and governance factors can affect stock price crash risk. These include the size of the local audit office (Callen, Fang, Xin, & Zhang, 2020), the nature of auditor-client relationship (Callen & Fang, 2017), the effectiveness of internal control systems (Chen, Chan, Dong, & Zhang, 2017), corporate board reforms (Hu, Li, Taboada, & Zhang, 2020), directors' external social networks (Fang, Pittman, & Zhao, 2021), and media coverage (An, Chen, Naiker, & Wang, 2020; Wu,

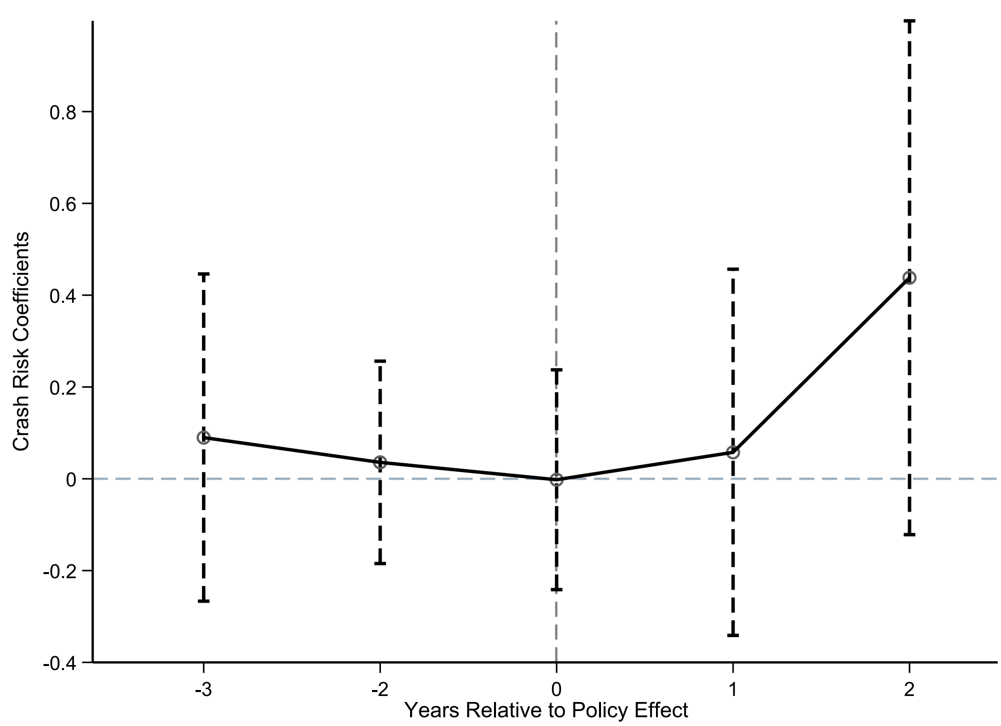

Fig. 1. The dynamic impact of KAM disclosures on the future stock price crash risk - *NCSKEW*.

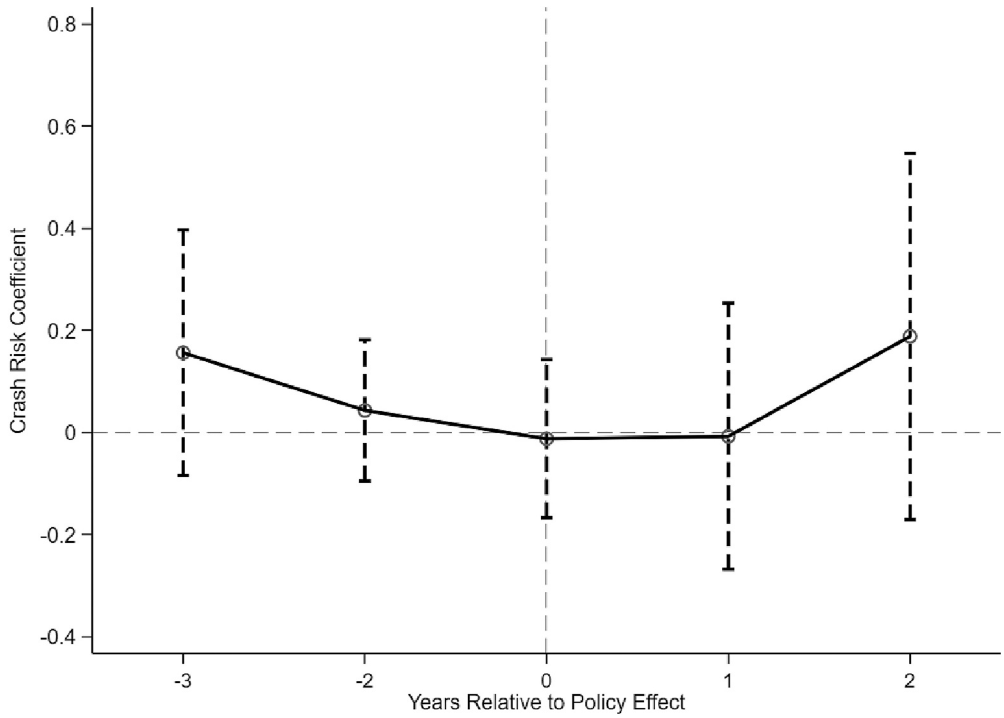

Fig. 2. The dynamic impact of KAM disclosures on the future stock price crash risk - *DUVOL*.

Table 6
Alternative measure of crash risk.

	(1)	(2)
Dependent variables =	CRASH	CRASH
	Coeff.	Coeff.
	(t-stat.)	(t-stat.)
DISKAM	**−0.031**	**−0.033**
	(−1.275)	**(−1.341)**
CONTROLS	No	Included
Firm and Year FE	Included	Included
Cluster Firm	Included	Included
N	18,751	18,751
Adj. R^2	0.018	0.024
F-value	25.641***	11.273***
Statistical power	1.000	1.000

Notes: This table presents the results of using alternative measure of crash risk, denoted by *CRASH*, which is an indicator variable that equals to one if a firm experienced at least one crash week within a given year, and zero otherwise. All other variables are defined in Appendix B. All continuous variables are winsorized at the top and bottom 1 %. All control variables are included in the regression but not tabulated for brevity. Robust t-statistics in parentheses. The reported significance is based on robust standard errors of two-tailed tests, adjusted for heteroscedasticity. *, **, and *** represent significance levels of 0.10, 0.05, and 0.01, respectively.

Xiong, Gao, & Zhang, 2022). Generally, these studies find that stronger monitoring and governance mechanisms help reduce stock price crash risk.

2.5. Hypotheses development

Our study aims to expand upon the existing research by delving deeper into the influence of audit-related factors on firm-specific stock price crash risk. The crash risk encompasses the occurrence of highly negative returns that has significantly impact not only on the performance of individual companies but also on the overall economy and the broader market (Callen & Fang, 2015a). Understanding and effectively managing the risk of stock price crashes is also essential for successful portfolio management (Hutton et al., 2009; Robin & Zhang, 2015). By examining the disclosures related to KAMs in relation to stock price crash risk, we can gain valuable insights into the factors that contribute to this risk and its implications for decision-making processes.

Regarding the role of the new expanded audit reports, the disclosure of KAMs aims to address the limitation of the traditional audit report, which has long been criticized for its simplistic pass-or-fail model that fails to effectively convey relevant information (Christensen, Glover, & Wolfe, 2014; Lennox et al., 2022). The Public Company Accounting Oversight Board (PCAOB) (2017) also argues that the traditional audit report does not adequately communicate critical information obtained and evaluated by auditors during the engagement. Under the new regulatory framework, auditors are now mandated to disclose significant audit challenges, risks, and judgments encountered during the audit to investors. Typically, these key or critical audit issues and matters were previously communicated exclusively to client audit committees and management. Thus, the disclosure of KAMs in the expanded audit report has the potential to improve corporate transparency by providing auditors with a means to directly communicate important private information to investors. However, it remains unclear whether the inclusion of KAMs would reduce managerial concealment of adverse news or significant risks, because there are two alternative schools of thought.

One school of thought aligns with the position that disclosure of KAMs can reduce corporate opacity by altering management behaviors indirectly. Managers have been shown to respond to KAM/CAM reporting by significantly changing financial statement disclosures and operating decisions (Bentley, Lambert, & Wang, 2021; Burke, Hoitash, Hoitash, & Xiao, 2022; Gold, Heilmann, Pott, & Rematzki, 2020; Tan & Yeo, 2022). Prior literature suggests that auditors have strong reputational and litigation risk incentives to prevent management from withholding bad news (Callen & Fang, 2017; Jensen & Meckling, 1976; Lys & Watts, 1994; Watts & Zimmerman, 1983). Since the reporting of KAMs potentially increases auditor liability and assessment of negligence (Gimbar et al., 2016; Vinson, Robertson, & Cockrell, 2019), we posit that mandatory disclosure of KAMs will incentivize auditors to pre-empt management from hoarding to reporting of significant risks. Because

Table 7
Regressions of stock price crash risk on adoption of KAM audit standard using alternative measurement horizons.

	(1)	(2)	(3)	(4)	(5)	(6)
	Seven-month Window		Eight-month Window		Nine-month Window	
Dependent Variable =	*NCSKEW*	*DUVOL*	*NCSKEW*	*DUVOL*	*NCSKEW*	*DUVOL*
DISKAM	**−0.099**	**−0.106**	**−0.084**	**−0.114**	**−0.081**	**−0.027**
	(−0.906)	**(−1.189)**	**(−0.735)**	**(−1.351)**	**(−0.700)**	**(−0.334)**
Controls	Included	Included	Included	Included	Included	Included
Firm and Year FE	Included	Included	Included	Included	Included	Included
Cluster Firm	Included	Included	Included	Included	Included	Included
N	16,889	16,889	16,889	16,889	16,889	16,889
Adj. R^2	0.038	0.039	0.040	0.044	0.050	0.053
F-value	14.669***	15.266***	14.681***	17.138***	16.417***	17.071***
Statistical power	1.000	1.000	1.000	1.000	1.000	1.000

Notes: This table shows the replicated results presented in Eq. (4) using alternative crash risk measurement windows. All variables are defined in Appendix B. All continuous variables are winsorized at the top and bottom 1 %. All control variables are included in the regression but not tabulated for brevity. Robust t-statistics in parentheses. The reported significance is based on robust standard errors of two-tailed tests, adjusted for heteroscedasticity. *, **, and *** represent significance levels of 0.10, 0.05, and 0.01, respectively.

such disclosures by management will help reduce information asymmetry between management and investors, the risk of a stock price crash is reduced.

Furthermore, in contrast to the traditional audit report with a simple pass/fail audit statement, the disclosure of KAMs is expected to expose firms to a higher level of scrutiny, reveal their risks, and ultimately improve corporate transparency. Increased corporate transparency can help mitigate information asymmetries and the divergence of investment opinions among investors by communicating critical private information through KAMs (Hong & Stein, 2003; Hutton et al., 2009; Robin & Zhang, 2015).[5] It can also enrich investors' knowledge of significant risks and areas of uncertainty surrounding corporate economic activities and underlying financial statements. Therefore, it reasons that the disclosure of KAMs should reduce stock price crash risk and we state the first hypothesis as below.

H1a. The disclosure of key audit matters in the expanded audit report is negatively associated with stock price crash risk.

An alternative school of thought suggests that the disclosure of KAMs does not reduce stock price crash risk. First, emerging research finds that the disclosure of KAMs in the new audit report contains little significant incremental information, suggesting that KAMs are not value relevant to the capital market (Files & Gencer, 2020; Gutierrez et al., 2018; Lennox et al., 2022; Liao et al., 2019). Second, since KAMs expose auditors to greater threat of litigation as documented by Gimbar et al. (2016) and Vinson et al. (2019), auditors are incentivized to reduce this threat. While not reporting KAMs can affect auditors' reputations if KAMs are expected or subsequently discovered, auditors may employ boilerplate KAM language that provides little or no new insight to investors (Brasel et al., 2016; International Auditing and Assurance Standards Board (IAASB), 2015; Public Company Accounting Oversight Board (PCAOB), 2017). By using standardized KAM language, auditors are able to minimize their litigation risk and decision effort (Rousseau & Zehms, 2020). Consequently, the expanded audit report containing KAMs may not reduce stock price crash risk. Taken together, the preceding competing discussions suggest that the impact of the expanded audit report on stock price crash risk remains an open empirical question. We state our second hypothesis as follows:

H1b. The disclosure of key audit matters in the expanded audit report is not related to stock price crash risk.

3. Sample selection and research design

3.1. Sample

We commence with all publicly traded Chinese firms listed on Shanghai and Shenzhen stock exchanges over the financial years 2012–2019. This period captures audited financial information up to four years before and three years after the financial year ending on December 31, 2016, the first year the disclosure of KAMs becomes mandatory for AH-share firms in China (25,328 observations).[6] We delete financial institutions (635 observations) and firms without market and financial data in the China Stock Market and Accounting Research (CSMAR) database (5942 observations). This sample selection procedure yields a final sample of 18,751 observations, including 76 unique AH-share firms with 536 firm-year observations and 3291 unique A-share firms yielding 18,215 firm-year observations.

3.2. Measure of firm-specific crash risk

Following Chen et al. (2001) and Kim et al. (2011a, 2011b), we use the negative coefficient of skewness of firm-specific weekly returns (*NCSKEW*) and the down-to-up volatility of firm-specific weekly returns (*DUVOL*) as the measures of stock price crash risk. For each firm-year, we use non-overlapping 12-month observations on skewness.[7] We first estimate firm-specific weekly returns, denoted by $W_{i,t}$ and operationalized as the natural logarithm value of one plus the residual return from Eq. (1), using the expanded market regression for each firm and year (Kim et al., 2011a, 2011b):

$$R_{j,t} = \alpha_j + \beta_{1j}R_{m,t-2} + \beta_{2j}R_{m,t-1} + \beta_{3j}R_{m,t} + \beta_{4j}R_{m,t+1} + \beta_{5j}R_{m,t+2} + \varepsilon_{j,t} \quad (1)$$

[5] In addition, Harris and Raviv (1993), Hong and Stein (2003), Hutton et al. (2009), Kandel and Pearson (1995), and Odean (1998) provide detailed discussion on the importance and consequences of differences of opinion among investors.

[6] In 2010, China Ministry of Finance requires all audit firms in China to reform their organizational structure from limited liability company (LLC) to limited liability partnership (LLP) before the end of 2011. Our sample begins in 2012 to avoid the confounding effect of changes in auditor's legal exposure (He, Pan, & Tian, 2017).

[7] Chen et al. (2001) argue that the choice of a horizon for crash risk measures is arbitrary. To reduce the measurement concerns on the 12-month returns as a proxy for stock crash, we also employ seven-month, eight-month, and nine-month horizons to measure crash risk in additional analyses and find consistent results.

Table 8
Mechanism tests.

	(1)	(2)
Dependent variable =	NCSKEW	DUVOL
Panel A: Information opaqueness - Analyst forecasts dispersion		
DISKAM	−0.066	−0.024
	(−0.544)	(−0.292)
DISKAM*FORECAST	**−0.038**	**−0.043**
	(−0.955)	**(−1.473)**
FORECAST	0.003	0.014
	(0.125)	(0.790)
Controls	Included	Included
Firm and Year FE	Included	Included
Cluster Firm	Included	Included
N	8205	8205
Adj. R^2	0.084	0.087
F-value	11.813***	11.872***
Statistical power	1.000	1.000
Panel B: Information opaqueness – Media coverage		
DISKAM	−0.087	−0.047
	(−0.809)	(−0.676)
DISKAM*MEDIA	**0.006**	**0.000**
	(0.172)	**(0.011)**
MEDIA	−0.038	−0.023
	(−1.468)	(−1.329)
Controls	Included	Included
Firm and Year FE	Included	Included
Cluster Firm	Included	Included
N	18,735	18,735
Adj. R^2	0.080	0.079
F-value	25.679***	25.951***
Statistical power	1.000	1.000
Panel C: Managerial opportunism		
DISKAM	−0.099	−0.047
	(−0.917)	(−0.676)
DISKAM*EM	**0.001**	**−0.010**
	(0.046)	**(−0.533)**
EM	−0.007	0.005
	(−0.341)	(0.402)
Controls	Included	Included
Firm and Year FE	Included	Included
Cluster Firm	Included	Included
N	18,575	18,575
Adj. R^2	0.080	0.078
F-value	25.292***	25.389***
Statistical power	1.000	1.000

Notes: This table presents the results of mechanism tests. *FORECAST* equals one if the analyst earnings forecasts dispersion is above the median value of sample observations and zero otherwise. *MEDIA* equals one if media coverage is below the median value of sample observations and zero otherwise. *EM* equals one if a firm's earnings management level is above the median value of sample observations and zero otherwise. All other variables are defined in Appendix B. All continuous variables are winsorized at the top and bottom 1 %. All control variables are included in the regression but not tabulated for brevity. Robust t-statistics in parentheses. The reported significance is based on robust standard errors of two-tailed tests, adjusted for heteroscedasticity. *, **, and *** represent significance levels of 0.10, 0.05, and 0.01, respectively.

where $R_{j,t}$ is the weekly return on stock j in week t, $R_{m,t}$ is the return on the value-weighted market index in week t. We include the lead and lag values of the value-weighted market indexes to control for non-synchronous trading (Dimson, 1979; Kim et al., 2011a). We estimate *NCSKEW* by taking the negative of the third moment of firm-specific weekly returns ($W_{i,t}$) for the corresponding period and dividing it by the standard deviation of the third moment of the firm-specific weekly returns:

$$NCSKEW_{j,t} = -\left[n(n-1)^{3/2}\sum W_{j,t}^3\right] \Big/ \left[(n-1)(n-2)\left(\sum W_{j,t}^2\right)^{3/2}\right] \tag{2}$$

where n is the number of trading weeks in period t. The negative sign in front of the third moment suggests that an increase in *NCSKEW* corresponds to a stock with a more left-skewed distribution (Chen et al., 2001).

To calculate the second stock price crash measure (*DUVOL*), we first divide all the sample firms into two groups based on the firm-specific weekly returns ($W_{i,t}$) over the 12-month period: "down" weeks with returns below the mean and "up" weeks with returns above the mean. We then compute the standard deviation for each of these subsamples, respectively. Finally, we take the natural logarithm of the ratio of the standard deviation on the down weeks to the standard deviation in the up weeks:

$$DUVOL_{j,t} = log\left\{\left[(n_u-1)\sum_{DOWN} W_{j,t}^2\right] \Big/ \left[(n_d-1)\sum_{UP} W_{j,t}^2\right]\right\} \tag{3}$$

where n_u and n_d are the number of weeks with returns above and below the average return in period t, respectively. A higher value of *DUVOL* indicates that a firm is more crash-prone.

3.3. Research design

The new auditor reporting requirements for AH-share firms in China took effect on January 1, 2017 and one year later for all A-share firms. We employ a staggered difference-in-differences (D-i-D) research design to capture the impact of KAM disclosure on stock price crash risk at different points in time. Our research approach is consistent with the existing literature that employs staggered difference-in-differences (D-i-D) methodology (e.g., Beck, Levine, & Levkov, 2010; Chan, Chen, Chen, & Yu, 2015; Cheng, De Franco, & Lin, 2022; Jiang, Wang, & Wang, 2019; Pan, Shroff, & Zhang, 2022), and requires to include firm and year fixed effects:

$$NCSKEW_{j,t} \Big/ DUVOL_{j,t} = \beta_0 + \beta_1 DISKAM_{j,t-1} + \beta_i \sum CONTROLS_{j,t-1} + FirmFE + YearFE + \varepsilon_{j,} \tag{4}$$

where, for firm j in year t, the dependent variable is the proxy for stock price crash risk, including *NCSKEW* and *DUVOL*.[8] The variable of interest, *DISKAM,* represents the mandated KAM disclosure and is a firm-year indicator that equals one if the firm is affected by the mandated KAM disclosure in those years when the new audit reporting standard is implemented, and zero otherwise. The coefficient on *DISKAM* captures the change in firm-level crash risk across pre- and post-adoption periods for a treatment firm compared to the change over the same sample period for a control firm. Specifically, the treatment firms are AH-share firms in financial years 2016, 2017, 2018 and 2019, and A-share firms in financial years 2017, 2018 and 2019, respectively.

Consistent with Kim et al. (2011a, 2011b), we include the lagged value of the following control variables that affect stock price crash risk. $NCSKEW_{t-1}$ is the lagged negative skewness of firm-specific weekly returns, and $DUVOL_{t-1}$ is the lagged value of the down-to-up volatility of the asymmetric volatilities between negative and positive firm-specific weekly returns. These two variables are used to control for the persistence of the dependent variables. $DTURN_{t-1}$ is the lagged value of the

[8] Following Hutton et al. (2009) and Kim et al. (2011a, 2011b), we also use an indicator variable to measure stock price crash risk. *CRASH* equals one if a firm experienced at least one crash week within a given year, and zero otherwise. Our results are robust to this alternative measure. Please see Section 4.3.2 on page 22 for the details.

Table 9
Heterogeneity analyses.

	(1)	(2)	(3)	(4)
Dependent variable =	*NCSKEW*	*DUVOL*	*NCSKEW*	*DUVOL*
Panel A: Corporate governance – board independence				
	Low Board Independence		High Board Independence	
DISKAM	**−0.047**	**−0.097**	**−0.047**	**−0.097**
	(−0.212)	**(−0.624)**	**(−0.212)**	**(−0.624)**
Controls	Included	Included	Included	Included
Firm and Year FE	Included	Included	Included	Included
Cluster Firm	Included	Included	Included	Included
N	9476	9476	9264	9264
Adj. R^2	0.103	0.097	0.103	0.097
F-value	14.232***	14.553***	12.812***	13.495***
Statistical power	1.000	1.000	1.000	1.000
Panel B: Corporate governance - institutional investors				
	Low Institutional Investors		High Institutional Investors	
DISKAM	**−0.085**	**−0.106**	**−0.085**	**−0.106**
	(−0.258)	**(−0.483)**	**(−0.258)**	**(−0.483)**
Controls	Included	Included	Included	Included
Firm and Year FE	Included	Included	Included	Included
Cluster Firm	Included	Included	Included	Included
N	9377	9377	9374	9374
Adj. R^2	0.091	0.098	0.091	0.098
F-value	17.904***	17.877***	14.427***	14.091***
Statistical power	1.000	1.000	1.000	1.000
Panel C: Product market competition				
	Strong Competition		Weak Competition	
DISKAM	**−0.040**	**−0.023**	**−0.169**	**−0.092**
	(−0.276)	**(−0.230)**	**(−0.980)**	**(−0.873)**
Controls	Included	Included	Included	Included
Firm and Year FE	Included	Included	Included	Included
Cluster Firm	Included	Included	Included	Included
N	10,556	10,556	8195	8195
Adj. R^2	0.083	0.089	0.083	0.078
F-value	17.686***	17.945***	9.923***	10.169***
Statistical power	1.000	1.000	1.000	1.000
Panel D: SOE and Non-SOE				
	SOE		Non-SOE	
DISKAM	**−0.148**	**−0.068**	**0.187**	**0.069**
	(−1.240)	**(−0.938)**	**(0.771)**	**(0.380)**
Controls	Included	Included	Included	Included
Firm and Year FE	Included	Included	Included	Included
Cluster Firm	Included	Included	Included	Included
N	7237	7237	11,514	11,514
Adj. R^2	0.081	0.078	0.075	0.073
F-value	11.601***	11.328***	20.306***	19.396***
Statistical power	1.000	1.000	1.000	1.000
Panel E: Big 6 and Non-Big 6 auditors				
	Big 6		Non-Big 6	
DISKAM	**−0.055**	**−0.027**	**−0.189**	**−0.079**
	(−0.416)	**(−0.326)**	**(−0.952)**	**(−0.584)**
Controls	Included	Included	Included	Included
Firm and Year FE	Included	Included	Included	Included
Cluster Firm	Included	Included	Included	Included
N	5543	5543	13,208	13,208
Adj. R^2	0.123	0.111	0.077	0.079
F-value	12.032***	11.038***	19.275***	20.814***
Statistical power	1.000	1.000	1.000	1.000

Notes: This table provides the results of the heterogeneity analyses. All variables are defined in Appendix B. All continuous variables are winsorized at the top and bottom 1 %. All control variables, and year and firm fixed effects are included in the regressions but not tabulated for brevity. Robust t-statistics addressing heteroscedasticity are reported in parentheses below the respective coefficient. *, **, and *** represent two-tailed significance levels of 0.10, 0.05, and 0.01, respectively.

average monthly stock turnover over the current financial year less the average monthly stock turnover over the previous financial year. The monthly stock turnover is calculated as the monthly trading volume divided by the total number of outstanding shares. $SIGMA_{t-1}$ is the standard deviation of firm-specific weekly returns in year *t-1*, which controls for stock volatility. RET_{t-1} is the mean of firm-specific weekly returns in year *t-1*, multiplied by 100. $SIZE_{t-1}$ is measured as the natural logarithm of total assets in year *t-1*. ROA_{t-1} is net income deflated by total assets in year *t-1*. $LOSS_{t-1}$ is coded one if the firm has incurred a loss in year *t-1* and zero otherwise. We also include REC_{t-1}, total receivables deflated by total assets in year *t-1*, and INV_{t-1}, total inventory deflated by total assets in year *t-1*, to control for financial reporting complexity.

LEV_{t-1} is the total liability deflated by total assets in year *t-1*. MB_{t-1} is the ratio of the market value of equity to the book value of equity in year *t-1*. In addition, we control for the quality of reporting using $DACC_{t-1}$, which is defined as the absolute value of discretionary accruals estimated from the modified Jones model (Hutton et al., 2009) in year *t-1*. SOE_{t-1} is coded as one if the firm is a state-owned enterprise in year *t-1* and zero otherwise. SHR_{t-1} is the percentage of shares held by the largest shareholders in year *t-1*. We also include $BIG6_{t-1}$, which is coded as one if the firm is audited by a Big 6 firm and zero otherwise. We further control for institutional shareholdings (INS_{t-1}), which is the percentage of shares held by institutional investors, and analyst followings (ANA_{t-1}), which is the natural logarithm of one plus the number of analyst followings. Yu (2008) and Desai and Dharmapala (2009) document that institutional investors and financial analysts play an important external monitoring role. Finally, we include firm and year fixed effects to capture any aggregate variations in firm-specific characteristics and years over time. Further, we cluster the standard errors at the firm level. All continuous variables are winsorized at the top and bottom 1 %. We perform regressions using robust standard errors corrected for firm-level heteroscedasticity. Appendix B defines the variables.

4. Empirical results

4.1. Descriptive statistics

Panel A of Table 1 presents the descriptive statistics for all the variables used in our main analyses. The mean and median values of *NCSKEW* are −0.286 and − 0.248, respectively. The mean and median values of *DUVOL* are −0.184 and − 0.185, respectively. The summary statistics of the stock price crash risk and control variables are comparable to the prior literature (DeFond, Hung, Li, & Li, 2015; Hutton et al., 2009).

Panels B presents the descriptive statistics of KAM content. Using textual analysis, we count the different KAM items, auditor responses, and the length of the respective disclosures. Panel B shows that, on average, firms report 2.084 KAMs, with a maximum of four KAMs. The average length of the KAM disclosures is approximately 415 words, with a minimum of 87 words and a maximum of 1131 words. Further, the average number of audit procedures to address KAMs is 11.250, and the average length of auditor responses is 594.300 words. On face value, these results suggest that auditors perform a relatively large number of procedures for a KAM and report the audit procedures performed.

Panel C reports the 10 most common KAM topics reported by auditors. The top five KAM topics relate to matters associated with revenue recognition (*REVENUE* = 31.064%), accounts receivable allowance and loans (*RECVLOAN* = 16.988%), intangible assets and goodwill (*INTANGIBLE* = 16.184%), inventory (*INVENTORY* = 10.395%), and property, plant, and equipment (*PPE* = 5.457%).

Table 2 provides the correlation matrix of the key variables and the correlation matrix of the KAMs content variables. We observe the correlations between the two measures of crash risk (*NCSKEW* and *DUVOL*) and *DISKAM*, and KAM content are less than the threshold of 0.5, indicating that multicollinearity is not a major concern.

4.2. Multivariate regression results

4.2.1. Expanded audit report and stock price crash risk

Table 3 presents the staggered D-i-D regression results testing our hypotheses. As a benchmark, we first report the results by only including our variable of interest (*DISKAM*), along with firm and year fixed effects in column (1) and (2). Results show that the coefficients on *DISKAM* are insignificant, whether based on the proxy of *NCSKEW* and *DUVOL* (t = −1.203 and − 1.114, respectively).

Further, the results in columns (3) and (4) show that *DISKAM* is insignificantly related to *NCSKEW* and *DUVOL* (t = −0.861 and − 0.731, respectively) after controlling for the potential firm-level predictors of crash risk. These findings suggest that, on average, the adoption of auditor KAM reporting is not associated with a change in firm-specific stock price crash risk between the periods before and after the adoption of KAM reporting requirements for both AH-share and A-share firms.[9]

In terms of the control variables, consistent with previous studies (e.g., Kim et al., 2011a, 2011b), firms that are larger in size (*SIZE*), have a larger standard deviation of firm-specific weekly returns (*SIGMA*), have greater return on assets (*ROA*) and market-to-book ratio (*MB*), and have incurred a loss (*LOSS*), are associated with higher levels of stock price crash risk, while firms that have higher financial leverage ratio (*LEV*) are associated with lower levels of stock price crash risk.

To alleviate the concern that we fail to detect significant effects, we estimate the power of the regressions using the Stata power command. As reported in Table 3, we have achieved a very high power of 100%, which suggests that our results are due to the absence of effects in the test variable rather than a failure to detect an effect. Taken together, these results are consistent with *H1b*.

4.2.2. KAM content and crash risk

To enrich our analyses on the impact of the disclosure of KAMs on firm-specific stock price crash risk and isolate the information content of KAMs from that of the annual report, we specifically examine whether the extent of KAM content and corresponding auditor responses have differential effects. Intuitively, greater KAM and auditor responses would seem more likely to convey additional new information to the capital markets and improve corporate transparency. For the KAM content, we examine the natural logarithm of the total number of KAMs (*KAMNUM*) and the natural logarithm of the total number of words describing KAMs (*KAMWRDS*). For auditor responses, we examine the natural logarithm of the total number of audit procedures employed to address KAMs (*RESPNUM*) and the natural logarithm of the total number of words in the auditor's responses to KAMs (*RESPWRDS*). Panel A of Table 4 shows that none of the coefficients on the KAM content variables are statistically significant in the stock-price crash risk regressions. For a more granular analysis, we examine if the top-5 KAM topics are related to stock-price crash risk. We focus on the top-5 because these are the most frequently occurring KAMs reported by auditors. Our results are similar across all other KAM topics. The results in Panel B of Table 4 consistently indicate a lack of significance on all KAM topic test variables. Again, the 100% statistical power of our regressions is very high suggesting that there indeed are no effects that we failed to detect.[10] Overall, these results reinforce our main conclusion that expanded audit reports in the form of KAM disclosures do not affect stock price crash risk.

Our results are inconsistent with regulatory intentions, which expect the adoption of the new expanded audit report to provide investors with more insightful information, and thus enhance corporate communication. However, our results are not surprising because empirical evidence on the consequences of the disclosure of KAMs is inconclusive and studies using the samples from the UK and the U.S. show that expanded audit report provides little incremental information to investors (Files & Gencer, 2020; Gutierrez et al., 2018; Lennox et al., 2022). Our research extends this line of studies by reporting unintended consequences of the disclosure of KAMs on stock price crash risk.

4.3. Robustness tests

4.3.1. Parallel trends assumption

To ensure that our design and analyses satisfy the parallel trends

[9] As a robustness test, we also control for industry fixed effects. Our results remain robust and the inferences are unchanged.

[10] In all subsequent regressions and reported in the tables, the statistical power is 100%.

assumption, we perform the following analysis. The D-i-D approach assumes that the treatment and control samples are similar in the outcome variable (i.e., stock price crash risk) in the absence of the new regulation (Abadie, 2005; Jiang et al., 2019; Roberts & Whited, 2013). We assess the validity of the parallel trends assumption underlying the D-i-D estimation by replacing *DISKAM* with a series of time indicator variables (i.e., *T-3, T-2,* $T+0$, $T+1$ and $T+2$) that track the year effect of the mandatory disclosure of KAMs, using year *T-1* as the benchmark. Specifically, *T-3* equals one if the financial year is up to three years before the mandatory disclosure of KAMs, and zero otherwise. *T-2* equals one if the financial year is two years before the mandatory disclosure of KAMs, and zero otherwise. *T* equals one for the financial year when the disclosure of KAMs is mandatory, and zero otherwise. $T+1$ is an indicator variable equal to one if the financial year is one year after the mandatory disclosure of KAMs, and zero otherwise. $T+2$ is an indicator variable equal to one if the financial year is two years after the mandatory disclosure of KAMs, and zero otherwise. The regression results reported in Table 5 show that the coefficients on the time indicator variables are statistically insignificant in the pre-KAMs period, suggesting that the treatment and control firms are similar in their stock-price crash risks in the absence of KAM regulation.

Following previous literature (e.g., Beck et al., 2010; Cheng et al., 2022; Pan et al., 2022), we also examine the parallel trends assumption by estimating the dynamic impact of KAM disclosures on future stock price crash risk before and after the year of implementation. Fig. 1 plots the coefficients by year using the regression specification in Column (1) of Table 5, while Fig. 2 plots the coefficients by year using the regression specification in Column (2) of Table 5. The dots (connected horizontally) represent the estimated coefficients, and the vertical lines represent 95% confidence intervals. Across both figures, we find no significant trend in crash risks during the pre-KAMs period.

4.3.2. Alternative measure of crash risk

Following Hutton et al. (2009) and Kim et al. (2011a, 2011b), we examine alternative measure of firm-level crash likelihood, denoted by *CRASH*, which is an indicator variable that equals to one if a firm experienced at least one crash week within a given year, and zero otherwise. Crash weeks are identified when the firm-specific weekly returns are 3.2 or more standard deviations below the mean. Consistent with Kim et al. (2011a, 2011b), the cutoff of 3.2 standard deviations is selected to generate a frequency of 0.1% in the normal distribution. The mean value of *CRASH* is 0.081, indicating that 8.1% of the samples experience at least one crash in a given year. The summary statistics of *CRASH* are comparable to the prior studies (Liang, Li, & Gao, 2020; Zhou, Li, Yan, & Lyu, 2021). As shown in Table 6, *DISKAM* is insignificantly related to *CRASH* both when excluding control variables ($t = -1.275$) and when including control variables ($t = -1.341$), suggesting that our results are robust to this alternative measure of stock price crash risk.

4.3.3. Crash horizon over alternative measure windows

In the main analyses, we use 12-month observations on skewness to measure future crash risk.[11] Chen et al. (2001) suggest using alternative horizon windows for crash risk measures for robustness. We use three alternative measurement intervals of crash risk, *NCSKEW* and *DUVOL,* over seven-, eight-, and nine-month horizons, and re-estimate the staggered D-i-D regression. The results reported in Table 7 are consistent with our main results and suggest that our results are not sensitive to different crash risk horizon windows.

5. Additional analyses

5.1. Mechanism tests

To better understand our results, we have performed two mechanism tests. Following prior literature, we examine potential channels including corporate information opaqueness (Andreou, Andreou, & Lambertides, 2021; Callen & Fang, 2015a), and managerial opportunism (Callen and Fang, 2015b; Kim et al., 2011a; Kim & Zhang, 2016).

5.1.1. Corporate information opaqueness

Prior literature finds that information opacity is a prominent factor resulting in extreme negative stock returns. For instance, using earnings management as a proxy for opacity of financial reports, Hutton et al. (2009) show that opaque firms are more likely to withhold bad news, which subsequently heightens stock price crashes. Similarly, Chen, Kim, and Yao (2017) document a positive association between earnings smoothing (a means to hide bad performance) and crash risk. Therefore, if the disclosure of KAMs reveals information on the potential downside risks and investors experiencing greater information asymmetry view the KAMs as new information, then such disclosure is expected to reduce information opaqueness.

Drawing on prior literature (An et al., 2020; Andreou et al., 2021; Callen & Fang, 2015a; Wu et al., 2022; Xu, Xuan, & Zheng, 2021), we use two indicators to measure corporate information opaqueness. The first measure is analyst earnings forecasts dispersion, which is defined as the standard deviation of earnings-per-share forecasts for the current fiscal year over the absolute value of the mean earnings forecasts. A higher value of forecasts dispersion indicates a lower level of transparency in firms' information environment. The second measure is firm-level media coverage, computed as the natural logarithm of one plus the total number of online financial media articles related to a firm. A greater level of media coverage represents a more transparent information environment. We obtain coverage data from the CNRDS database.

To test this mechanism, following Balachandran, Duong, Luong, and Nguyen (2020)'s approach, we first create two dummy variables (*FORECAST*, *MEDIA*). *FORECAST* equals one if the analyst earnings forecasts dispersion is above the median value of sample observations and zero otherwise. *MEDIA* equals one if media coverage is below the median value of sample observations and zero otherwise. We then interact *DISKAM* with *FORECAST* or *MEDIA* and include the interaction term in our staggered D-i-D model. The interaction term captures the effect of the new auditing standards on stock price crash risk conditional on corporate information opaqueness.

The results are presented in Panels A and B of Table 8. With the coefficients on *DISKAM* remain insignificant, the interaction term *DISKAM*FORECAST* and *DISKAM*MEDIA* are insignificantly related to *NCSKEW* and *DUVOL*, suggesting that the information role of the new auditing reporting is not significant in firms with higher level of information opaqueness.

5.1.2. Managerial opportunism

Firms with a higher level of earnings management are more likely to hide information from the capital market, which would cause stock price crashes. Prior studies have demonstrated that opportunistic managerial behavior is a mechanism by which their variables of interest affect firm-specific stock price crash risk (e.g., Andreou et al., 2021; Kim et al., 2011a). In light of these studies, we seek to investigate the mediating effect of managerial opportunism.

Following prior literature (Callen & Fang, 2015a; Kim et al., 2011a; Kim & Zhang, 2016), we use earnings management as a proxy for managerial opportunistic behaviors. Specifically, we measure earnings management by summing up the absolute value of discretionary accruals in the prior three years, where discretionary accruals are

[11] The crash risk measures we employ are estimated over the 12-month period ending four months after the financial year end (i.e., May 1 in year t to April 30 in year t + 1) to account for the effect of the release of the annual report, because all listed firms in China are required to file the annual reports before April 30. Our use of a 12-month window is consistent with the prior literature (Callen & Fang, 2015b; Kim et al., 2011a; Kim et al., 2016).

estimated from the modified Jones model. We then generate a dummy variable, denoted as *EM*, which equals one if a firm's earnings management level is above the median value of sample observations and zero otherwise. In the mechanism analysis, we add the interaction term *DISKAM**EM* in the regression model to capture the effects of KAM disclosure on crash risk conditional on managerial opportunism. In Panel C of Table 8, the coefficients of *DISKAM**EM* are not significant, indicating that there is no difference between firms with higher levels of earnings management level and firms less engaged in earnings management in terms of the monitoring effects of new auditing standards on bad news hoarding behavior.

5.2. Heterogeneity analyses

To further examine whether the insignificant results are consistent across different subgroups of the sample, we have performed a number of cross-sectional analyses. Prior research has identified several factors associated with crash risk, including corporate governance (An & Zhang, 2013; Callen & Fang, 2015b), market competition (Kim & Zhang, 2016; Li & Zhan, 2019), ownership structure (Piotroski, Wong, & Zhang, 2015), and auditor size (Callen et al., 2020; Robin & Zhang, 2015). In this section, we investigate whether our results are affected by these factors. Overall, as we document below, we find that our primary findings based on the two measures of stock price crash risk are not sensitive, but are robust to these potential influences. The regressions reported in Table 9 include all the control variables and fixed effects but are not tabulated to conserve space.

5.2.1. Corporate governance

Prior studies report that strong corporate governance can deter managerial misconduct and constrain managers' opportunistic behaviors (e.g., Beasley, 1996; Carcello, Hermanson, & Ye, 2011). An and Zhang (2013) and Callen and Fang (2013) find that firms with better corporate governance are less prone to crash risk because monitoring by governance systems can alleviate agency risks and attenuate bad news hoarding. Sila, Gonzalez, and Hagendorff (2017) show that firms with more independent directors voluntarily disclose more information and display lower crash risk. Similarly, Andreou, Antoniou, Horton, and Louca (2016) argue that a higher proportion of independent directors reduces the likelihood of a crash. Accordingly, the strength of corporate governance could affect our results, whereby the expanded audit report may be more likely to reduce crash risk for firms with poor corporate governance.

To test this effect, we employ the percentage of independent directors serving on the board and institutional ownership as measures for corporate governance. We partition the full sample into strong and weak corporate governance based on the median value of these two governance variables. The results in Panels A and B of Table 9 indicate that the coefficients on *DISKAM* remain insignificant across firms with more and less independent directors and institutional investors. This suggests that the strength of corporate governance does not affect our main findings.

5.2.2. Product market competition

According to the proprietary cost hypothesis and information concealing theory, firms operating in competitive markets have incentives to hoard bad news (Li & Zhan, 2019), which could affect our results. To investigate the impact of market competition on our main results, we partition the sample into strong and weak product market competition subsamples based on the median values of the Herfindahl-Hirschman index (Hoberg & Phillips, 2010). This index is computed as the sum of squared market shares for firms in each industry for each year. Consistent with Callen and Fang (2015a), we use firms' assets to compute a firm's market share. When we re-estimate Eq. (4), the results in Panel C of Table 9 again show that the coefficient on *DISKAM* is statistically insignificant. This analysis indicates that our results are not affected by product market competition.

5.2.3. Ownership structure

About fifty 50% of Chinese listed firms are SOEs that gives the government significant voting rights in these firms (Firth, Fung, & Rui, 2006). Piotroski et al. (2015) document that ownership structure affects stock price crash risk. Compared to non-SOEs, SOEs typically have more opaque financial reporting (Bushman, Piotroski, & Smith, 2004), and thus may be more likely to suppress negative information because of political incentives (Bushman & Piotroski, 2006). To test the effect of SOEs on our main results, we re-estimate Eq. (4) for subsamples that are SOEs and non-SOEs. The results presented in Panel D show that the coefficient on *DISKAM* in all four columns remains statistically insignificant, suggesting that corporate ownership structure does not affect the association between KAMs and stock-price crash risk. Thus, our primary results are robust to ownership structure.

5.2.4. Auditor size

We consider whether the size of the audit firm affects our main results, since investors associate higher audit quality with large audit firms (DeAngelo, 1981; Eshleman & Guo, 2014). Panel E in Table 9 re-estimates Eq. (4) separately for subsamples audited by the Big 6 firms and non-Big 6 firms.[12] The insignificant coefficient on *DISKAM* suggest that the impact of the expanded audit report on firm-level stock price crash risk is not influenced by the size of the audit firm.

6. Conclusions

In this study, we examine the economic consequences of the new mandatory auditor reporting standards that have been implemented around the world. We base our study in China because it has adopted KAMs auditor reporting standard that permits analyzing a longer horizon; the PCAOB's implementation of the equivalent standard in the U.S. is recent and does not permit undertaking a longer event-window analysis. Using a staggered D-i-D design, we find that the adoption of the new expanded audit report is not significantly related to firm-specific stock price crash risk. Textual analysis shows that the extent of KAM content and auditor responses to these audit matters are not associated with stock price crash risk. In the mechanism tests, we do not find that the disclosure of KAMs is associated with corporate information opaqueness and managerial opportunism. We further show that the baseline results do not vary across different levels of corporate governance, product market competition, ownership structure, or the size of the audit firm. Our regression results are robust and achieve very high statistical power that gives us confidence to conclude that the adoption of the KAM auditor reporting standard and the KAMs disclosed by auditors do not significantly affect the capital market.

Our study adds to the emerging literature on the economic consequences of the adoption of the new auditor reporting standards in China. Our results inform standard-setters and regulators around the world, including the IAASB, FRC, MOF, and PCAOB who are seeking empirical evidence on the impact of the unprecedented expanded audit report on investors, the primary users of audited financial information. Our findings imply that mandating auditors to disclose KAMs has not had the intended effect of improving corporate transparency. Holding all else constant, our robust findings suggest that this new disclosure does not increase transparency and informativeness, amidst the backdrop of arguments against the costs of providing this disclosure and exposing auditors to potential litigation.

While our results are generalizable and can inform stakeholders around the world because China follows the ISA 701 standard on auditor reporting of KAM, we suggest caution due to differences in audit institutional frameworks and capital market characteristics. Furthermore, we recognize that the early results of our study, along with those of

[12] We also partition the samples into subsamples audited by the Big 4 firms and non-Big 4 firms and find consistent results.

concurrent studies, should be interpreted in context of the capital market still familiarizing with and comprehending the new KAM disclosures. Future research can consider capital market learning effects and whether investors find KAM disclosures to be value relevant.

Declaration of Competing Interest

No.

Data availability

Data will be made available on request.

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Appendix A. Example of key audit matters

Air China (00753) – Audited by KPMG

Assessing the carrying value of aircraft and flight equipment	
Refer to note 15 to the consolidated financial statements and the accounting policies on page 109	
The key audit matter	How the matter was addressed in our audit Our audit procedures to assess the carrying value of aircraft and flight equipment included the following:
The carrying value of aircraft and flight equipment is reviewed annually taking into consideration factors such as changes in fleet composition, current and forecast market values and technical factors which may have a material impact on any impairment charges for the year. We identified the assessment of the carrying value of aircraft and flight equipment as a key audit matter because of its significance to the consolidated financial statements and because assessing the key assumptions underlying impairment assessments involves a significant degree of judgment by management in considering the nature, timing and likelihood of changes to the factors noted above which may affect both the carrying value of the Group's aircraft and flight equipment as well as any impairment charges for the year	• discussing with strategy and development department the plans for future fleet composition including future acquisitions and retirement of aircraft as well as indicators of possible impairment of aircraft and flight equipment and, where such indicators were identified, assessing whether management had performed impairment assessments in accordance with the requirements of the prevailing accounting standards; • assessing management's assertions and estimates adopted in their impairment assessments with reference to valuation reports published by third party specialists, our knowledge of the airline industry and the Group's historical experience and future operating plans; • challenging the assumptions and critical judgments adopted by management by comparing management's past estimates and plans to the current year's estimates and plans and taking into account recent developments in the airline industry and future operating plans.

Appendix B. Variable definitions

Variable Name	Operational Definition
Dependent variables	
NCSKEW	The negative skewness of firm-specific weekly returns, calculated by taking the negative of the third moment of firm-specific weekly returns for the corresponding period and dividing it by the standard deviation of the third moment of the firm-specific weekly returns.
DUVOL	The "down-to-up" volatility measure of the asymmetric volatilities between negative and positive firm-specific weekly returns.
CRASH	An indicator variable equal to one if a firm experienced at least one crash week within a given year, and zero otherwise.
Variables of interest	
DISKAM	An indicator variable equal to one if the audit report contains a KAM, and zero otherwise.
Control variables	
$NCSKEW_{t-1}$	The negative skewness of firm-specific weekly returns in the previous financial year.
$DUVOL_{t-1}$	The "down-to-up" volatility of the asymmetric volatilities between negative and positive firm-specific weekly returns in the previous financial year.
$DTURN_{t-1}$	The lag value of the average monthly stock turnover over the current financial year less the average monthly stock turnover over the previous financial year.
$SIGMA_{t-1}$	The standard deviation of firm-specific weekly returns in the previous financial year.
RET_{t-1}	The mean of firm-specific weekly returns in the previous financial year, times 100.
$SIZE_{t-1}$	The natural logarithm of total assets in the previous financial year.
ROA_{t-1}	Net income deflated by total assets in the previous financial year.
$LOSS_{t-1}$	An indicator variable equal to one if the firm has incurred a loss in the previous financial year, and zero otherwise.
REC_{t-1}	Total receivables deflated by total assets in the previous financial year.
INV_{t-1}	Total inventory deflated by total assets in the previous financial year.
LEV_{t-1}	Total liabilities deflated by total assets in the previous financial year.
MB_{t-1}	Market value of equity to the book value of equity in the previous financial year.

(*continued on next page*)

(*continued*)

Variable Name	Operational Definition
$DACC_{t-1}$	Absolute value of discretionary accruals estimated from the modified Jones model in the previous financial year.
SOE_{t-1}	An indicator variable equal to one if the firm is a state-owned enterprise in the previous financial year, and zero otherwise.
SHR_{t-1}	The percentage of shares held by the largest shareholder in the previous financial year.
$BIG6_{t-1}$	An indicator variable equal to one if the firm is audited by a Big 6 audit firm in the previous financial year, and zero otherwise.
INS_{t-1}	The percentage of shares held by institutional investors in the previous financial year.
ANA_{t-1}	The natural logarithm value of one plus the number of analyst followings in the previous financial year.
$FORECAST_{t-1}$	A dummy variable equal to one if the analyst earnings forecasts dispersion is above the median value of sample observations, and zero otherwise.
$MEDIA_{t-1}$	A dummy variable equal to one if media coverage is below the median value of sample observations, and zero otherwise.
EM_{t-1}	A dummy variable equal to one if a firm's earnings management level is above the median value of sample observations, and zero otherwise.
KAM content	
KAMNUM	The natural logarithm of the total number of KAMs.
KAMWRDS	The natural logarithm of the total number of words describing KAMs.
RESPNUM	The natural logarithm of the total number of audit procedures employed to address KAMs.
RESPWRDS	The natural logarithm of the total number of words in the auditor's responses to KAMs.
REVENUE	An indicator variable equal to one if the auditor reports revenue recognition as a KAM, and 0 otherwise.
RECVLOAN	An indicator variable equal to one if the auditor reports accounts receivable allowance and impairment of loans as a KAM, and 0 otherwise.
INTANGIBLE	An indicator variable equal to one if the auditor reports impairment of intangible assets and goodwill as a KAM, and 0 otherwise.
INVENTORY	An indicator variable equal to one if the auditor reports inventory recognition as a KAM, and 0 otherwise.
PPE	An indicator variable equal to one if the auditor reports property, plant and equipment measurement and impairment as a KAM, and 0 otherwise.

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